Infectious Diseases II: Endocarditis, SSTI/Osteomyelitis, CNS, Intra-abdominal, C. difficile

All key points for the exam (AHA, ESC, IDSA, SIS, ACG) — drug table after each topic

PART A — INFECTIVE ENDOCARDITIS (IE)

1. Pathogenesis, Microbiology and Clinical Features

- Infection of the **endocardium** (usually **heart valves**) → **vegetation** (platelets, fibrin, bacteria, inflammatory cells) • bacteria inside vegetations are **dense, slow-growing, protected** → need **bactericidal** drugs, **high doses**, **IV**, and **prolonged duration** (4–6 weeks).
- **Risk factors: prosthetic valve**, **prior IE**, congenital heart disease, rheumatic/degenerative valve disease, **injection drug use (IDU)**, intravascular devices (pacemaker/ICD, catheters), hemodialysis, poor dentition, diabetes, elderly.
- **Acute IE** (days; virulent — **S. aureus**) vs **subacute** (weeks–months; **viridans streptococci**, enterococci).
- **Native valve (NVE)** vs **prosthetic valve (PVE)** — early PVE (< 1 year: S. aureus, **coagulase-negative staphylococci**, gram-negatives, fungi) vs late PVE (> 1 year: like NVE).
- **IDU → right-sided (tricuspid)** IE, **S. aureus** (MRSA) → septic pulmonary emboli.

Organism	Typical setting
Staphylococcus aureus (MSSA/MRSA)	**Most common overall** (~30%) • acute • IDU, healthcare-associated, intravascular devices, PVE • high mortality, abscesses
Viridans group streptococci	Subacute NVE • **dental** procedures/poor dentition
Streptococcus gallolyticus (S. bovis)	Elderly • **associated with colon cancer → colonoscopy**
Enterococci (E. faecalis)	Elderly men • **GU/GI** procedures • intrinsically resistant to cephalosporins
Coagulase-negative staphylococci (S. epidermidis)	**Early prosthetic valve**, devices
HACEK (Haemophilus, Aggregatibacter, Cardiobacterium, Eikenella, Kingella)	Slow-growing gram-negatives • culture-negative-like • large vegetations
Culture-negative	Prior antibiotics (most common cause), **Bartonella**, **Coxiella burnetii** (Q fever), **Brucella**, Tropheryma whipplei, fungi
Fungi (Candida, Aspergillus)	IDU, PVE, long-term catheters, TPN, immunosuppression → **surgery required**

Clinical features

- **Fever** (most common, ~90%), **new or changing heart murmur**, malaise, weight loss, night sweats, arthralgia, splenomegaly, anemia, ↑ ESR/CRP.
- **Embolic (vascular) phenomena:** stroke, septic pulmonary emboli (right-sided), renal/splenic infarcts, **mycotic aneurysms**, **Janeway lesions** (**painless** hemorrhagic macules on palms/soles), conjunctival hemorrhages, **splinter hemorrhages**.
- **Immunologic phenomena: Osler nodes** (**painful** nodules on finger/toe pads), **Roth spots** (retinal hemorrhages with pale centers), **glomerulonephritis**, positive rheumatoid factor.
- **Complications: heart failure** (valve destruction — most common cause of death), **perivalvular abscess** (new heart block), emboli (stroke), metastatic infection (vertebral osteomyelitis, septic arthritis, splenic abscess), AKI.

Diagnosis

- **Blood cultures: ≥ 3 sets** from separate venipuncture sites (≥ 1 h apart if stable) **before antibiotics** • bacteremia is continuous.
- **Echocardiography: TTE first** (quick) • **TEE** if TTE negative but suspicion high, **prosthetic valve**, intracardiac device, S. aureus bacteremia, or suspected complications (abscess) — more sensitive • cardiac CT and **^{18}F-FDG PET/CT** (PVE, devices).
- **Modified Duke criteria (2023 Duke-ISCVID):**
- **Major:** (1) **microbiologic** — typical organism in ≥ 2 separate blood cultures (viridans strep, S. gallolyticus, HACEK, **S. aureus**, E. faecalis, CoNS/others in PVE) or persistent bacteremia, or **Coxiella** serology (phase I IgG > 1:800) • (2) **imaging** — vegetation, abscess, perforation, new valvular regurgitation, new prosthetic dehiscence (echo/CT/PET) • (3) surgical/pathologic findings.
- **Minor: predisposition** (prior IE, prosthetic valve, IDU, congenital/valve disease), **fever ≥ 38 °C**, **vascular phenomena**, **immunologic phenomena**, microbiologic evidence not meeting major criteria.
- **Definite IE: 2 major**, or **1 major + 3 minor**, or **5 minor** • possible: 1 major + 1 minor, or 3 minor.

2. Treatment of Infective Endocarditis (AHA 2015 / ESC 2023)

- **Duration counts from the first negative blood culture** (if initial cultures positive) • repeat blood cultures every 24–48 h until negative.
- **Gentamicin synergy** (with cell-wall agents for streptococci/enterococci): **3 mg/kg/day** (once daily for streptococci; divided q8h for enterococci) • target **peak 3–4 mcg/mL, trough < 1** • monitor renal function and hearing.
- **Vancomycin**: AUC 400–600 (or trough 15–20 historically) • **daptomycin** 8–10 mg/kg (higher than label 6 mg/kg) for S. aureus/enterococcal IE.
- **Outpatient parenteral therapy (OPAT)** once stable • **partial oral step-down** (POET trial; ESC 2023) acceptable in selected stable patients with left-sided IE (streptococci, E. faecalis, MSSA, CoNS) after ≥ 10 days of IV therapy and no abscess.

Streptococcal IE (viridans group streptococci, S. gallolyticus)

Susceptibility	Native valve (NVE)	Prosthetic valve (PVE)
Highly penicillin-susceptible (MIC ≤ 0.12)	**Aqueous penicillin G 12–18 million units/day IV** (q4h or continuous) **or ceftriaxone 2 g IV daily × 4 weeks • or 2-week regimen**: penicillin G or ceftriaxone **+ gentamicin 3 mg/kg daily × 2 weeks** (uncomplicated, normal renal function, no abscess) • PCN allergy: **vancomycin × 4 weeks**	Penicillin G 24 MU/day or ceftriaxone × **6 weeks** ± gentamicin first 2 weeks • vancomycin × 6 weeks if allergic
Relatively resistant (MIC > 0.12 to < 0.5)	**Penicillin G 24 MU/day × 4 weeks + gentamicin × first 2 weeks** (ceftriaxone alone if susceptible)	Penicillin G or ceftriaxone × 6 weeks **+ gentamicin × 6 weeks**
Resistant (MIC ≥ 0.5)	Treat as **enterococcal IE**	Treat as enterococcal IE

Staphylococcal IE

Organism	Native valve (NVE)	Prosthetic valve (PVE)
MSSA	**Nafcillin or oxacillin 12 g/day IV (2 g q4h) × 6 weeks • cefazolin 2 g q8h × 6 weeks** if non-anaphylactic PCN allergy (cefazolin increasingly preferred) • **gentamicin NOT recommended** for NVE (nephrotoxicity, no benefit) • uncomplicated right-sided in IDU: 2 weeks may suffice	**Nafcillin/oxacillin + rifampin 300 mg PO/IV q8h ≥ 6 weeks + gentamicin × first 2 weeks**
MRSA (or severe PCN allergy)	**Vancomycin × 6 weeks** (AUC 400–600) **or daptomycin ≥ 8 mg/kg/day × 6 weeks**	**Vancomycin + rifampin ≥ 6 weeks + gentamicin × first 2 weeks**

- **Rifampin** in PVE (penetrates **biofilm** on prosthetic material) — start **after 3–5 days** of effective therapy when bacteremia cleared (avoid resistance) • DDIs.
- **S. aureus bacteremia** (any): always **TEE**, ID consult, remove infected catheters, repeat cultures • **β-lactam is superior to vancomycin for MSSA** → switch as soon as MSSA known.

- **Daptomycin** is not for **left-sided** IE monotherapy with low doses • **never for pneumonia** • check **CPK weekly**.

Enterococcal IE (E. faecalis)

- **Ampicillin 2 g IV q4h + ceftriaxone 2 g IV q12h × 6 weeks** (**double β-lactam**; preferred if **CrCl < 50**, **high-level aminoglycoside resistance**, or to avoid nephrotoxicity) • **or ampicillin (or penicillin G) + gentamicin 1 mg/kg q8h × 4–6 weeks** (4 weeks NVE with symptoms < 3 months; 6 weeks for PVE or longer symptoms).
- **Penicillin allergy:** desensitize, or **vancomycin + gentamicin × 6 weeks**.
- **VRE** (E. faecium): **daptomycin 10–12 mg/kg** ± ampicillin/ceftaroline, or **linezolid** • surgery often needed.
- **Cephalosporins alone have NO activity** against enterococci (ceftriaxone works only synergistically with ampicillin).

Other organisms

- **HACEK: ceftriaxone 2 g daily** (or ampicillin-sulbactam, ciprofloxacin) × **4 weeks** NVE / 6 weeks PVE.
- **Culture-negative (empiric):** acute NVE → **vancomycin + cefepime** • subacute NVE → **vancomycin + ampicillin-sulbactam** • PVE (< 1 year) → **vancomycin + gentamicin + cefepime + rifampin** • Bartonella → doxycycline + gentamicin • Coxiella → doxycycline + hydroxychloroquine ≥ 18 months.
- **Fungal IE: valve surgery** + **liposomal amphotericin B ± flucytosine** or high-dose echinocandin → lifelong azole suppression.
- **Brucella IE:** aminoglycoside + doxycycline + rifampin + surgery.

Surgery and other management

- **Indications for early surgery: heart failure** (severe regurgitation, obstruction) • **uncontrolled infection** (perivalvular **abscess**, heart block, fistula, **persistent bacteremia > 5–7 days** despite therapy, **fungal** or highly resistant organisms) • **prevention of embolism** (vegetation **> 10 mm** with embolic event, or very large > 30 mm) • most early PVE with S. aureus.
- Remove infected **pacemaker/ICD** (complete device extraction).
- **Antithrombotics: do not start aspirin** or anticoagulation for IE itself • consider holding anticoagulation in S. aureus PVE with stroke (ICH risk).
- **S. gallolyticus** → colonoscopy • **IDU** → addiction treatment (MOUD).

IE prophylaxis (AHA 2021)

- **Only for highest-risk patients**: (1) **prosthetic heart valve** or prosthetic material used for valve repair (incl. TAVR, annuloplasty rings, chords), (2) **previous IE**, (3) **unrepaired cyanotic congenital heart disease** or repaired CHD with residual shunts/valvular regurgitation adjacent to prosthetic material, (4) **repaired CHD with prosthetic material within 6 months**, (5) **cardiac transplant with valvulopathy**.
- **Only for dental procedures** involving **manipulation of gingival tissue** or the periapical region of teeth or **perforation of the oral mucosa** (extractions, cleaning) • **NOT** for routine anesthetic injections through non-infected tissue, X-rays, orthodontic adjustments, GI/GU procedures, or tattoos.
- **Good oral hygiene** is more important than prophylaxis.

Situation	Agent (single dose 30–60 min before procedure)	Adult / child
Oral	**Amoxicillin**	**2 g** / 50 mg/kg
Unable to take oral	Ampicillin IM/IV • or cefazolin or ceftriaxone IM/IV	2 g / 50 mg/kg • 1 g / 50 mg/kg
Penicillin allergy — oral	Cephalexin (if no anaphylaxis/angioedema/urticaria history) • **azithromycin or clarithromycin** • **doxycycline**	2 g / 50 mg/kg • **500 mg** / 15 mg/kg • 100 mg / 2.2 mg/kg
Penicillin allergy — unable to take oral	Cefazolin or ceftriaxone (non-anaphylactic)	1 g / 50 mg/kg

- **Clindamycin is no longer recommended** for IE prophylaxis (2021 — C. difficile risk).
- If the dose was missed, can give up to **2 h after** the procedure.

3. Endocarditis Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Penicillin G aqueous	12–24 million units/day IV (q4h or continuous)	Hypersensitivity, seizures (high doses), hyperkalemia (K salt)	Strep IE • renal adjust
Ceftriaxone	**2 g IV daily** (enterococcus 2 g q12h)	Biliary sludge, C. diff	Once daily (OPAT) • no enterococcal activity alone
Ampicillin	**2 g IV q4h**	Rash, diarrhea	Enterococcus (+ ceftriaxone or gentamicin)
Nafcillin / oxacillin	**2 g IV q4h** (12 g/day)	**Phlebitis**, neutropenia, **interstitial nephritis**, hypokalemia, hepatitis (oxacillin)	**Nafcillin = CYP3A4 inducer** → ↓ warfarin • no renal adjustment
Cefazolin	**2 g IV q8h**	Rash	MSSA (non-anaphylactic PCN allergy) • renal adjust
Vancomycin	15–20 mg/kg q8–12h • load 25–35 mg/kg • **AUC 400–600**	**Nephrotoxicity**, **vancomycin infusion reaction** (infuse ≥ 1 h), ototoxicity, neutropenia	MRSA • AKI ↑ with pip-tazo, aminoglycosides
Daptomycin	**8–10 mg/kg IV daily** (VRE 10–12)	**Myopathy/rhabdomyolysis** (↑ CPK), **eosinophilic pneumonia**	**CPK weekly** • hold statins • **not for pneumonia** • renal adjust (q48h CrCl < 30)
Gentamicin (synergy)	**3 mg/kg/day** (strep: once daily • enterococcus: 1 mg/kg q8h)	**Nephrotoxicity, ototoxicity**	Peak 3–4, trough < 1 • not for MSSA NVE
Rifampin	**300 mg q8h** (PVE staph)	Hepatitis, orange fluids	Biofilm • **potent inducer** (warfarin, DOACs) • start after bacteremia clears
Linezolid	600 mg q12h	Thrombocytopenia, serotonin syndrome	VRE
Amoxicillin (prophylaxis)	**2 g PO** 30–60 min before dental procedure	—	Highest-risk patients only
Azithromycin (prophylaxis)	500 mg PO	QT	PCN allergy • clindamycin no longer recommended

PART B — SKIN AND SOFT TISSUE INFECTIONS (SSTI) AND OSTEOMYELITIS

4. Skin and Soft Tissue Infections (IDSA 2014)

- **Main organisms: Streptococcus pyogenes (GAS) → non-purulent** (cellulitis, erysipelas) • **Staphylococcus aureus (incl. MRSA) → purulent** (abscess, furuncle, carbuncle) • polymicrobial in diabetic foot, bites, necrotizing infections.
- **Severity: mild** (no systemic signs) • **moderate** (systemic signs: fever > 38, HR > 90, RR > 24, WBC > 12,000 or < 4,000) • **severe** (failed oral therapy, SIRS + hypotension, immunocompromised, deep infection/necrotizing signs).
- **MRSA risk factors:** prior MRSA, recent hospitalization/antibiotics, IDU, HIV, MSM, incarceration, military, athletes (contact sports), dialysis, purulence.

Impetigo and ecthyma

- Superficial infection (children) — **honey-colored crusts** (non-bullous) or bullae (bullous — S. aureus) • S. aureus and/or GAS.

- **Limited: topical mupirocin 2% BID × 5 days** (or retapamulin, ozenoxacin) • **numerous lesions/outbreaks/ecthyma:** oral **cephalexin** or **dicloxacillin** × 7 days • MRSA → **doxycycline**, **clindamycin**, **TMP-SMX** • GAS only → penicillin.

Purulent SSTI (abscess, furuncle, carbuncle)

Severity	Management
Mild	**Incision and drainage (I&D) alone** (primary therapy) • warm compresses
Moderate (systemic signs)	**I&D + oral antibiotics covering MRSA**: **TMP-SMX DS 1–2 tabs BID** or **doxycycline 100 mg BID** (clindamycin alternative) • 5–10 days • adding TMP-SMX/clindamycin to I&D improves cure even for uncomplicated abscesses
Severe	**I&D + IV MRSA coverage: vancomycin**, **daptomycin** 4 mg/kg, **linezolid** 600 mg q12h, telavancin, **ceftaroline**, dalbavancin/oritavancin (single/two-dose long-acting) • MSSA → **nafcillin, cefazolin**, then cephalexin/dicloxacillin
Recurrent abscesses	Culture • decolonization: **intranasal mupirocin BID × 5 days** + **chlorhexidine washes** + decontaminate personal items

Non-purulent SSTI (cellulitis, erysipelas)

- **Erysipelas**: superficial, **raised, sharply demarcated**, bright red (face, legs) — **GAS** • **cellulitis**: deeper, **poorly demarcated** erythema, warmth, edema, pain.
- Risk: **tinea pedis**/skin breaks, lymphedema, venous insufficiency, obesity, prior cellulitis • usually **no cultures** needed (blood cultures if severe/immunocompromised).

Severity	Management
Mild	**Oral anti-streptococcal**: **penicillin VK 500 mg QID**, **cephalexin 500 mg QID**, **dicloxacillin 500 mg QID**, or **clindamycin 300–450 mg TID** (PCN allergy) • **5 days** (extend if not improved)
Moderate	**IV**: **penicillin G**, **ceftriaxone 1–2 g daily**, **cefazolin 1–2 g q8h**, or clindamycin 600 mg q8h • add MRSA coverage if penetrating trauma, IDU, MRSA colonization/infection elsewhere, purulent drainage, or SIRS
Severe	**Vancomycin + piperacillin-tazobactam** (or carbapenem) • **urgent surgical evaluation** to rule out necrotizing infection

- **Elevate the affected limb**, treat predisposing conditions (**tinea pedis** → topical antifungal, edema, skin hydration) • margins may expand in the first 24–48 h even with appropriate therapy.
- **Recurrent cellulitis** (3–4 episodes/year despite risk-factor control): **prophylactic penicillin V 250 mg BID** or erythromycin, or **benzathine penicillin IM every 2–4 weeks**.
- Systemic corticosteroids (prednisone 40 mg × 7 days) may be considered in non-diabetic adults.

Necrotizing soft tissue infections (necrotizing fasciitis)

- **Surgical emergency → prompt surgical debridement** (most important) + broad-spectrum antibiotics.
- **Clues: pain out of proportion** to exam, rapid progression, **tense edema**, bullae, skin necrosis/anesthesia, **crepitus (gas)**, systemic toxicity, ↑ CK, hyponatremia (LRINEC score).
- **Type I** (polymicrobial — diabetes, Fournier's gangrene) • **Type II** (monomicrobial — **GAS**, S. aureus) • Clostridium (gas gangrene) • **Vibrio vulnificus** (seawater, raw oysters, liver disease) • Aeromonas (freshwater).
- **Empiric: vancomycin (or linezolid) + piperacillin-tazobactam** (or carbapenem, or ceftriaxone + metronidazole) • **add clindamycin** for toxin suppression.
- **GAS or Clostridium confirmed: penicillin G + clindamycin** (clindamycin inhibits **toxin production** — Eagle effect) • ± IVIG for streptococcal toxic shock.
- **Vibrio vulnificus: doxycycline + ceftazidime** (or ceftriaxone) • **Aeromonas:** doxycycline + ciprofloxacin or ceftriaxone.

Animal and human bites

- **Dog/cat: Pasteurella multocida** (cat bites — deep puncture, high infection rate), Capnocytophaga canimorsus (asplenia), S. aureus, streptococci, anaerobes • **human: Eikenella corrodens**, streptococci, S. aureus, anaerobes (clenched-fist injury).

- **Treatment/preemptive (3–5 days)** for high-risk wounds (**cat bites**, hand/face/genital wounds, deep punctures, crush injuries, immunocompromised, asplenia, edema): **amoxicillin-clavulanate 875/125 mg BID** (drug of choice) • IV: ampicillin-sulbactam, pip-tazo, carbapenem.
- **Penicillin allergy: doxycycline**, or **TMP-SMX / cefuroxime / FQ + anaerobic coverage (metronidazole or clindamycin)**, moxifloxacin alone • **avoid cephalexin, dicloxacillin, clindamycin, erythromycin alone** (miss Pasteurella).
- **Tetanus** prophylaxis • **rabies** assessment (bats, wild animals, unvaccinated dogs) • irrigation; avoid primary closure (except face).

Diabetic foot infection (IDSA/IWGDF 2023)

- Diagnosis is **clinical** (≥ 2 signs of inflammation or purulence) • don't culture/treat clinically uninfected ulcers • **deep tissue culture** after debridement (not swabs).
- **Mild** (superficial, cellulitis ≤ 2 cm): **aerobic gram-positive cocci** (S. aureus, streptococci) → **cephalexin**, dicloxacillin, **amoxicillin-clavulanate**, clindamycin • MRSA risk → **doxycycline** or **TMP-SMX** • **1–2 weeks**.
- **Moderate–severe** (deeper, > 2 cm, systemic signs): **polymicrobial** (gram-positive, gram-negative, anaerobes) → **amoxicillin-clavulanate**, **ampicillin-sulbactam**, **ceftriaxone + metronidazole**, **ertapenem**, FQ + metronidazole • **Pseudomonas** coverage (**pip-tazo**, cefepime) only if risk (macerated wound, warm climate, prior isolation, water exposure) • **add vancomycin/linezolid if MRSA risk** • **2–4 weeks** (soft tissue).
- **Osteomyelitis** (probe-to-bone positive, X-ray/MRI, ↑ ESR) → bone biopsy culture → **6 weeks** without resection • **~3 weeks after minor amputation** with positive bone margin • a few days if all infected bone removed.
- Also: **offloading**, debridement, revascularization (PAD), glycemic control, surgery for abscess/gangrene.

Other SSTIs

- **Surgical site infection:** open the incision, drain • antibiotics if systemic signs or > 5 cm erythema • prevention: **cefazolin 2 g (3 g if ≥ 120 kg) within 60 min before incision**, redose every 4 h, stop within 24 h • vancomycin/fluoroquinolones within 120 min.
- **Pressure ulcers/burns:** local care; systemic antibiotics only for cellulitis/sepsis • burns: topical silver sulfadiazine (sulfa allergy, leukopenia), mafenide (metabolic acidosis).
- **Erythrasma, folliculitis** (hot tub → Pseudomonas, self-limited).

5. Osteomyelitis and Septic Arthritis

- **Osteomyelitis** = bone infection • **hematogenous** (children — long bones; adults — **vertebral**), **contiguous** (trauma, surgery, prosthetic joints, pressure ulcers), **vascular insufficiency** (diabetic foot) • acute (< 2 weeks) vs chronic (sequestrum, sinus tract).
- **Organisms: S. aureus** (most common overall) • CoNS (prosthetic material) • streptococci • **Salmonella** (**sickle cell disease**) • **Pseudomonas** (**nail puncture through sneaker**, IDU) • gram-negatives (elderly, UTI → vertebral) • **Brucella** and **TB (Pott's)** in vertebral osteomyelitis (endemic areas) • Kingella kingae (young children) • polymicrobial (diabetic foot).
- **Diagnosis:** localized pain, swelling, fever • ↑ **ESR/CRP** (monitor response) • **MRI** (most sensitive imaging) • X-ray changes late (10–14 days) • **bone biopsy/culture** = gold standard (hold antibiotics if stable) • blood cultures (vertebral).
- **Treatment: 6 weeks** (IV or highly bioavailable oral — **OVIVA trial**: oral non-inferior) + **surgical debridement** of necrotic bone (chronic).
- **MSSA: nafcillin/oxacillin 2 g q4h** or **cefazolin 2 g q8h** (ceftriaxone alternative) • **MRSA: vancomycin** or **daptomycin 6–8 mg/kg** (linezolid, TMP-SMX + rifampin alternatives) • **Pseudomonas:** cefepime, ceftazidime, or **ciprofloxacin** (oral, excellent bone penetration) • streptococci: penicillin G, ceftriaxone • gram-negatives: ceftriaxone or FQ.
- **Oral options** (good bone levels): **FQs**, **linezolid**, **TMP-SMX**, **clindamycin**, **doxycycline**, **rifampin (combination only)**, metronidazole.
- **Vertebral osteomyelitis (IDSA 2015):** blood cultures + biopsy (hold antibiotics if neurologically stable) • **6 weeks** • surgery for neurologic deficit/abscess/instability.
- **Prosthetic joint infection:** debridement and implant retention (early) with **rifampin-based combination for staphylococci** (biofilm) or one-/two-stage exchange • 3–6 months.

- **Septic arthritis:** acute monoarticular hot swollen joint (knee) → **arthrocentesis** (WBC > 50,000, neutrophils, Gram stain/culture) + drainage • **S. aureus** most common → **vancomycin** (± ceftriaxone if gram-negative risk) • **gonococcal** (young sexually active; migratory arthritis, tenosynovitis, pustules) → **ceftriaxone 1 g IV daily** (+ test/treat chlamydia) • duration 2–4 weeks.

6. SSTI and Bone Infection Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Mupirocin 2%	Topical BID–TID × 5 d • nasal BID × 5 d (decolonization)	Local irritation	Impetigo • MRSA decolonization
Cephalexin	**500 mg QID**	GI, rash	Non-purulent cellulitis, MSSA • no MRSA
Dicloxacillin	500 mg QID (empty stomach)	GI, hepatitis	MSSA, strep
Penicillin VK	500 mg QID • recurrence prophylaxis 250 mg BID	Rash	Strep cellulitis/erysipelas
Clindamycin	300–450 mg PO TID • 600–900 mg IV q8h	**C. difficile (boxed)**, diarrhea, rash	**Toxin suppression** (GAS, necrotizing, TSS) • D-test for inducible resistance
TMP-SMX DS	**1–2 tabs BID**	Hyperkalemia, rash/SJS, marrow	**MRSA** (poor strep coverage) • ↑ INR
Doxycycline	**100 mg BID**	Photosensitivity, esophagitis	MRSA, bites • chelation • avoid pregnancy, < 8 years
Amox-clavulanate	**875/125 mg BID**	Diarrhea	**Bites**, diabetic foot (mild–moderate)
Cefazolin	1–2 g IV q8h (surgical prophylaxis 2 g; 3 g if ≥ 120 kg)	Rash	MSSA, cellulitis, surgical prophylaxis
Nafcillin / oxacillin	1–2 g IV q4h	Phlebitis, interstitial nephritis, neutropenia	MSSA • nafcillin ↓ warfarin
Vancomycin	15–20 mg/kg q8–12h (AUC 400–600)	Nephrotoxicity, infusion reaction	MRSA
Linezolid	**600 mg PO/IV q12h**	**Thrombocytopenia**, serotonin syndrome, neuropathy	Avoid SSRIs, tyramine • 100% oral bioavailability
Daptomycin	SSTI 4 mg/kg • bacteremia/bone 6–8 mg/kg daily	**CPK ↑/myopathy**, eosinophilic pneumonia	Weekly CPK • hold statins
Ceftaroline	600 mg IV q12h	Rash, Coombs+	MRSA-active cephalosporin
Dalbavancin / oritavancin	Dalba 1,500 mg once (or 1 g + 500 mg day 8) • orita 1,200 mg once	Infusion reactions, ↑ LFTs	Long-acting, OPAT/IDU • oritavancin ↑ aPTT/INR falsely
Pip-tazobactam	4.5 g IV q6–8h	AKI with vanco, thrombocytopenia	Severe/necrotizing, diabetic foot (Pseudomonas)
Ciprofloxacin	750 mg PO BID (bone)	FQ class	Pseudomonas osteomyelitis • chelation
Ertapenem	1 g daily	Seizures	Diabetic foot • no Pseudomonas
Rifampin	300–450 mg BID (combination only)	Orange fluids, hepatitis	Biofilm/hardware • inducer

PART C — CENTRAL NERVOUS SYSTEM (CNS) INFECTIONS

7. Bacterial Meningitis

- **Classic triad: fever, neck stiffness (nuchal rigidity), altered mental status** (all 3 in < 50%) • **headache** (most common), photophobia, vomiting, seizures, **Kernig's** and **Brudzinski's** signs • **petechial/purpuric rash → N. meningitidis** • infants: irritability, poor feeding, **bulging fontanelle**.
- **Medical emergency** → start antibiotics **within 1 hour • do not delay antibiotics** for CT or LP.
- **CT before LP** only if: **immunocompromised**, history of CNS disease (mass, stroke), **new-onset seizure** (within 1 week), **papilledema**, **abnormal level of consciousness**, **focal neurologic deficit** → sequence: **blood cultures → dexamethasone + empiric antibiotics → CT → LP**.

CSF finding	Normal	Bacterial	Viral	Fungal / TB
Opening pressure	< 20 cm H2O	**↑**	Normal/mild ↑	**↑ (crypto very high)**
WBC (cells/mm³)	< 5	**1,000–5,000** (↑↑)	50–1,000	20–500
Predominant cell	Lymphocytes	**Neutrophils (PMNs)**	**Lymphocytes**	Lymphocytes
Glucose	45–80 mg/dL (CSF:serum ≈ 0.6)	**↓ (< 40; ratio < 0.4)**	**Normal**	**↓**
Protein	15–45 mg/dL	**↑↑ (100–500)**	Normal/mild ↑	↑
Gram stain/other	—	Positive in 60–90%	PCR (HSV, enterovirus)	India ink, **cryptococcal antigen**, AFB

Empiric therapy by age and risk (IDSA)

Patient	Likely pathogens	Empiric regimen
< 1 month	**Group B Streptococcus** (S. agalactiae), **E. coli**, **Listeria monocytogenes**, Klebsiella	**Ampicillin + cefotaxime** (or cefepime) **or ampicillin + gentamicin • avoid ceftriaxone in neonates** (bilirubin displacement/kernicterus; Ca precipitates)
1–23 months	**S. pneumoniae**, **N. meningitidis**, GBS, H. influenzae, E. coli	**Vancomycin + ceftriaxone (or cefotaxime)**
2–50 years	**N. meningitidis**, **S. pneumoniae**	**Vancomycin + ceftriaxone (or cefotaxime)**
> 50 years, alcoholism, **immunocompromised**, pregnancy	S. pneumoniae, N. meningitidis, **Listeria**, aerobic gram-negatives	**Vancomycin + ceftriaxone + ampicillin** (Listeria — **cephalosporins have NO Listeria activity**)
Post-neurosurgery, CSF shunt, penetrating head trauma	**S. aureus**, **CoNS**, aerobic gram-negatives incl. **Pseudomonas**, Acinetobacter	**Vancomycin + cefepime** (or ceftazidime or **meropenem**)
Basilar skull fracture	S. pneumoniae, H. influenzae, GAS	Vancomycin + ceftriaxone
Severe β-lactam allergy	—	Vancomycin + **moxifloxacin** (or aztreonam/meropenem per allergy) + **TMP-SMX** (Listeria)

Meningitic doses (high, to penetrate CSF)

- **Ceftriaxone 2 g IV q12h • cefotaxime 2 g q4–6h • vancomycin 15–20 mg/kg q8–12h** (trough 15–20 / AUC) • **ampicillin 2 g q4h • cefepime 2 g q8h • meropenem 2 g q8h • penicillin G 4 million units q4h • acyclovir 10 mg/kg q8h** • gentamicin 5 mg/kg/day • TMP-SMX 5 mg/kg q6–12h.
- CSF penetration ↑ with **inflamed meninges** • β-lactams need high doses • **aminoglycosides and 1st/2nd-generation cephalosporins penetrate poorly** (not used) • clindamycin, daptomycin, echinocandins poor CNS penetration.

Adjunctive dexamethasone

- **Dexamethasone 0.15 mg/kg IV q6h × 2–4 days** (adults 10 mg q6h × 4 days) — give **10–20 min before or with the first antibiotic dose** (not after).
- **Benefit:** ↓ **neurologic sequelae/hearing loss** and mortality in **adults with S. pneumoniae**, and **H. influenzae b in children**.
- **Stop** if the organism is **not S. pneumoniae** (adults) / not Hib (children) • do not give if antibiotics already started • not in neonates.
- May ↓ **vancomycin CSF penetration** (↓ inflammation) — some add **rifampin** if dexamethasone used with resistant pneumococcus.

Pathogen-directed therapy and duration

Organism	Preferred therapy	Duration
N. meningitidis	**Penicillin G** or ampicillin (if PCN MIC < 0.1) • **ceftriaxone/cefotaxime**	**7 days**
H. influenzae	Ampicillin (β-lactamase negative) • **ceftriaxone** (β-lactamase positive)	**7 days**
S. pneumoniae	PCN MIC < 0.06 → penicillin G/ampicillin • ceftriaxone MIC < 1 → ceftriaxone • MIC ≥ 1 → **vancomycin + ceftriaxone** (± rifampin)	**10–14 days**
Group B Streptococcus	Ampicillin or penicillin G (± gentamicin)	**14–21 days**
Listeria monocytogenes	**Ampicillin or penicillin G ± gentamicin** • PCN allergy → **TMP-SMX**	**≥ 21 days**
Aerobic gram-negative bacilli (E. coli, Klebsiella)	Ceftriaxone/cefotaxime • Pseudomonas → cefepime, ceftazidime, meropenem	**21 days**
S. aureus	MSSA → nafcillin/oxacillin • MRSA → vancomycin (± rifampin), linezolid alternative	≥ 14 days

Infection control and chemoprophylaxis

- **N. meningitidis** → **droplet precautions** for 24 h after effective therapy • **chemoprophylaxis for close contacts** (household, daycare, direct exposure to oral secretions — kissing, intubation without mask) **within 24 h**: **rifampin 600 mg PO q12h × 2 days** (children 10 mg/kg q12h) • **ciprofloxacin 500 mg PO single dose** (adults) • **ceftriaxone 250 mg IM single dose** (125 mg < 15 years; **preferred in pregnancy**) • azithromycin 500 mg single dose (where cipro resistance) • the patient also needs eradication if treated with penicillin (not needed after ceftriaxone).
- **H. influenzae type b**: **rifampin 20 mg/kg (max 600 mg) daily × 4 days** for all household contacts if any household member is < 4 years and not fully vaccinated, or immunocompromised child.
- **S. pneumoniae**: no chemoprophylaxis • vaccinate.
- **Vaccines**: Hib, PCV/PPSV, MenACWY, MenB (see immunizations).

8. Other CNS Infections

- **Viral (aseptic) meningitis: enteroviruses** (most common), HSV-2, VZV, HIV, arboviruses → supportive (acyclovir if HSV/VZV suspected).
- **HSV encephalitis** (HSV-1, **temporal lobe**): fever, **altered behavior/personality**, seizures, focal deficits • CSF lymphocytes + RBCs, **HSV PCR**, MRI temporal lobe changes • **start acyclovir 10 mg/kg IV q8h empirically** when encephalitis suspected → **14–21 days** • dose on **ideal body weight** (obese) • **ADRs: crystal nephropathy (AKI)** → **hydrate**, renal adjust • **neurotoxicity** (confusion, tremor) • phlebitis.
- **Brain abscess:** streptococci, anaerobes, S. aureus, gram-negatives • **ceftriaxone/cefotaxime + metronidazole** (± vancomycin if S. aureus/post-surgical) • **6–8 weeks** + aspiration/drainage.
- **Cryptococcal meningitis** (HIV, CD4 < 100): induction **liposomal amphotericin B + flucytosine × 2 weeks** (or WHO single high-dose L-AmB 10 mg/kg + flucytosine + fluconazole) → consolidation **fluconazole 400–800 mg × 8 weeks** → maintenance 200 mg ≥ 1 year • **serial LPs for ↑ ICP** (no acetazolamide, steroids) • **delay ART 4–6 weeks** (IRIS).

- **TB meningitis:** RIPE **9–12 months** + **dexamethasone** • **neurosyphilis: aqueous penicillin G 18–24 million units/day IV × 10–14 days** • **Lyme meningitis:** ceftriaxone or doxycycline 14–21 days • **neurobrucellosis:** ceftriaxone + doxy + rifampin.
- **Healthcare-associated ventriculitis/shunt infection:** vancomycin + cefepime/meropenem • remove device • **intraventricular** antibiotics (vancomycin, aminoglycosides) if poor response (preservative-free).

9. CNS Infections Drug Summary

Drug	Meningitic dose	Key ADRs	Key notes / interactions
Ceftriaxone	**2 g IV q12h**	Biliary sludge, C. diff	**Avoid in neonates** (bilirubin, Ca precipitates) • no Listeria activity
Cefotaxime	**2 g IV q4–6h**	Rash, C. diff	Neonates (if available)
Vancomycin	**15–20 mg/kg q8–12h** (trough 15–20)	Nephrotoxicity, infusion reaction	Resistant pneumococcus, MRSA • dexamethasone ↓ penetration
Ampicillin	**2 g IV q4h**	Rash, seizures (high dose)	**Listeria** (< 1 month, > 50 years, immunocompromised)
Penicillin G	4 million units q4h (24 MU/day)	Seizures, hypersensitivity	Meningococcus, susceptible pneumococcus, neurosyphilis
Cefepime	**2 g IV q8h**	**Neurotoxicity** (renal failure)	Post-neurosurgical, Pseudomonas
Meropenem	**2 g IV q8h**	Seizures (low), ↓ valproate	Post-neurosurgical, β-lactam alternative
Dexamethasone	**0.15 mg/kg (10 mg) q6h × 2–4 d**, before/with 1st antibiotic	Hyperglycemia, GI bleeding	Stop if not pneumococcus/Hib
Acyclovir	**10 mg/kg IV q8h** (IBW) × 14–21 d	**Crystal nephropathy**, neurotoxicity, phlebitis	**Hydrate** • renal adjust • HSV/VZV
Rifampin (prophylaxis)	Meningococcus 600 mg q12h × 2 d • Hib 20 mg/kg (600) daily × 4 d	Orange fluids	**↓ OCs** (backup) • DDIs • not in pregnancy (use ceftriaxone)
Ciprofloxacin (prophylaxis)	500 mg PO once	FQ effects	Adults (meningococcus)
Ceftriaxone (prophylaxis)	250 mg IM once (125 mg < 15 y)	Injection pain	**Pregnancy**
TMP-SMX	5 mg/kg (TMP) q6–12h IV	Rash, marrow, hyperkalemia	Listeria in PCN allergy
Metronidazole	500 mg IV q8h	Neuropathy, metallic taste	Brain abscess (anaerobes)

PART D — INTRA-ABDOMINAL INFECTIONS (IAI)

10. Intra-abdominal Infections (IDSA/SIS)

- **Types: primary peritonitis** (spontaneous bacterial peritonitis — no GI perforation; cirrhosis/ascites) • **secondary peritonitis** (GI perforation, appendicitis, diverticulitis, postoperative, trauma) • **tertiary** (persistent/recurrent after adequate therapy; resistant organisms, Candida) • **intra-abdominal abscess** • biliary (cholecystitis, cholangitis) • **peritoneal dialysis peritonitis**.
- **Uncomplicated** (confined to one organ) vs **complicated** (extends into peritoneal space → peritonitis/abscess).
- **Microbiology (secondary): polymicrobial** — **Enterobacterales** (**E. coli**, Klebsiella), **anaerobes** (**Bacteroides fragilis**), streptococci, enterococci (healthcare-associated/postoperative), Pseudomonas and Candida (healthcare-associated).
- **SBP:** usually **monomicrobial** — E. coli, Klebsiella, streptococci (ascitic PMN ≥ 250 cells/mm³).

- **Symptoms:** abdominal pain, fever, guarding/rigidity, rebound tenderness, ileus, N/V • CT scan for diagnosis.
- **Management = source control (drainage, debridement, repair) within 24 h** + **fluid resuscitation** + **antibiotics** (start within 1 h if septic shock).

Empiric antibiotic regimens

Setting	Regimens
Community-acquired, mild–moderate (e.g., perforated appendix, abscess)	**Single agent: ertapenem 1 g daily**, moxifloxacin 400 mg (resistance concerns) • **Combination: cefazolin, cefuroxime, ceftriaxone, or cefotaxime + metronidazole • ciprofloxacin or levofloxacin + metronidazole** (if local E. coli FQ susceptibility > 90%)
Community-acquired, high risk/severity (APACHE II > 15, septic shock, immunocompromised, delayed source control)	**Piperacillin-tazobactam 4.5 g q6h • meropenem 1 g q8h**, **imipenem-cilastatin 500 mg q6h • cefepime or ceftazidime + metronidazole** • (FQ + metronidazole alternatives)
Healthcare-associated (postoperative, recent antibiotics)	Based on local resistance: **meropenem/imipenem**, **pip-tazo**, cefepime/ceftazidime + metronidazole ± **vancomycin** (MRSA risk) ± **enterococcal coverage** (ampicillin/vancomycin; VRE in liver transplant) ± aminoglycoside • **antifungal (echinocandin)** if Candida isolated and severe/healthcare-associated • newer agents for MDR (ceftolozane-tazobactam + metronidazole, ceftazidime-avibactam + metronidazole, eravacycline)

- **Not recommended** empirically: **ampicillin-sulbactam** (high E. coli resistance), **cefotetan and clindamycin** (**B. fragilis resistance**), **aminoglycosides** routinely (toxicity; alternatives exist), **empiric antifungal/enterococcal coverage in community-acquired** infections, tigecycline as first-line.
- **Anaerobic coverage** needed for distal small bowel, appendix, and colon sources (and proximal GI with obstruction): **metronidazole**, carbapenems, β-lactam/β-lactamase inhibitors, moxifloxacin, tigecycline/eravacycline.
- **Duration: 4 days (fixed) after adequate source control** (**STOP-IT trial**) • longer if source control inadequate • **uncomplicated appendicitis removed surgically: ≤ 24 h** (or none postop) • bowel injury repaired within 12 h: ≤ 24 h • acute stomach/proximal jejunal perforations (no acid suppression) repaired within 24 h: 24 h.
- Switch to oral (e.g., amoxicillin-clavulanate, cipro/levo + metronidazole, cefuroxime + metronidazole) when tolerating diet.

Specific intra-abdominal conditions

- **Appendicitis:** appendectomy (standard) • **antibiotics-alone** option for uncomplicated appendicitis (CODA trial; ~30% eventually need surgery) • perioperative cefoxitin/cefazolin + metronidazole.
- **Diverticulitis: uncomplicated** in immunocompetent patients → **antibiotics may be omitted** (selective use — AGA 2021) • when used: amoxicillin-clavulanate, or ciprofloxacin/TMP-SMX + metronidazole × 4–7 days • **complicated** (abscess, perforation) → IV antibiotics + percutaneous drainage/surgery.
- **Acute cholecystitis/cholangitis (Tokyo guidelines):** mild–moderate → ceftriaxone + metronidazole, ertapenem, FQ + metronidazole, ampicillin-sulbactam (if local E. coli susceptible) • severe/healthcare-associated → pip-tazo, cefepime + metronidazole, carbapenem ± vancomycin • **cholecystectomy / biliary drainage (ERCP)** • 4–7 days after source control.
- **Spontaneous bacterial peritonitis: cefotaxime 2 g IV q8h** or **ceftriaxone 2 g daily** × **5–7 days** + **IV albumin 1.5 g/kg day 1 and 1 g/kg day 3** (if SCr > 1, BUN > 30, or bilirubin > 4) • nosocomial → pip-tazo/carbapenem • **secondary prophylaxis**: **ciprofloxacin 500 mg daily** or **TMP-SMX DS daily** (norfloxacin not available in US) • **GI bleed in cirrhosis: ceftriaxone 1 g daily × 7 days**.
- **Necrotizing pancreatitis: no prophylactic antibiotics** • infected necrosis → **carbapenem** (or FQ + metronidazole) + delayed drainage/necrosectomy.
- **Peritoneal dialysis peritonitis** (cloudy effluent, WBC > 100, PMN > 50%): **intraperitoneal vancomycin or cefazolin** (gram-positive) **+ ceftazidime/cefepime or aminoglycoside** (gram-negative) • ≥ 2–3 weeks.

- **Liver abscess:** pyogenic (Klebsiella) → drainage + ceftriaxone + metronidazole 4–6 weeks • **amebic** (Entamoeba histolytica) → **metronidazole** then luminal agent (paromomycin).
- **H. pylori**, gastroenteritis: see GI topics.

11. Intra-abdominal Infections Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Metronidazole	**500 mg IV/PO q8–12h**	**Metallic taste**, nausea, **peripheral neuropathy** (long courses), seizures, dark urine	**Disulfiram-like reaction with alcohol** (labeled) • **↑ warfarin INR** (2C9) • ↑ lithium • QT • 100% oral bioavailability
Ceftriaxone	1–2 g daily (SBP 2 g)	Biliary sludge, C. diff	+ metronidazole for anaerobes
Cefazolin / cefuroxime	2 g q8h / 1.5 g q8h	Rash	+ metronidazole (mild CA-IAI)
Cefotaxime	**2 g q8h** (SBP)	Rash	SBP
Ertapenem	**1 g IV daily**	Seizures, C. diff	**No Pseudomonas, Acinetobacter, enterococci** • ESBL • **↓ valproate**
Meropenem / imipenem	1 g q8h / 500 mg q6h	Seizures (**imipenem** > meropenem)	Severe/HA-IAI • ↓ valproate • renal adjust
Pip-tazobactam	4.5 g q6h (or 3.375 g q8h extended)	AKI with vancomycin, thrombocytopenia	Covers anaerobes + Pseudomonas + enterococcus (E. faecalis)
Cefepime / ceftazidime	2 g q8h	Neurotoxicity (cefepime)	+ metronidazole
Ciprofloxacin / levofloxacin	400 mg IV q12h (500–750 PO) / 750 mg daily	FQ class	+ metronidazole • SBP prophylaxis cipro 500 mg daily
Moxifloxacin	400 mg daily	QT, hepatotoxicity	Single agent (mild) • B. fragilis resistance rising
Amox-clavulanate	875/125 mg BID	Diarrhea	Oral step-down, diverticulitis
Eravacycline / tigecycline	1 mg/kg q12h / 100 mg → 50 mg q12h	N/V, tooth discoloration • **tigecycline boxed: ↑ mortality**	MDR • no Pseudomonas • poor serum levels (not bacteremia)
Ceftolozane-tazo / ceftazidime-avi	1.5–3 g q8h / 2.5 g q8h (+ metronidazole)	GI, C. diff	MDR Pseudomonas / KPC Enterobacterales
Micafungin / caspofungin	100 mg daily / 70 → 50 mg	Hepatic, infusion reactions	Candida in HA-IAI
Albumin 25% (SBP)	**1.5 g/kg day 1, 1 g/kg day 3**	Volume overload	↓ hepatorenal syndrome and mortality

PART E — CLOSTRIDIOIDES DIFFICILE INFECTION (CDI)

12. C. difficile Infection (IDSA/SHEA 2021, ACG 2021)

- **Gram-positive, spore-forming anaerobic bacillus → toxins A and B** → colitis (pseudomembranous colitis) • **spores** survive on surfaces and are **not killed by alcohol-based hand sanitizer → soap and water** handwashing + **contact precautions** (gown, gloves) + **sporicidal (bleach) cleaning**.
- **Risk factors: antibiotics** (most important) — highest risk: **clindamycin**, **fluoroquinolones**, **cephalosporins** (3rd/4th generation), **carbapenems**, broad-spectrum penicillins (any antibiotic can cause it, even single doses) • **age > 65**, **hospitalization/long-term care**, **PPIs** (and H2RAs), chemotherapy, immunosuppression, IBD, GI surgery, tube feeds, prior CDI.
- **Hypervirulent strain** NAP1/BI/027 (↑ toxin production, FQ resistance).

- **Symptoms: ≥ 3 unformed (watery) stools in 24 h**, abdominal cramping, fever, **leukocytosis** (can be very high), ↑ SCr • severe: ileus, toxic megacolon, perforation.

Diagnosis

- **Test only patients with ≥ 3 unformed stools in 24 h** (new and unexplained) • **test only unformed stool • do not test** if on laxatives within 48 h.
- **Do NOT test asymptomatic patients** (colonization common) and **do NOT do a "test of cure"** (toxin/PCR may stay positive).
- **Tests: NAAT (PCR)** for toxin genes (very sensitive — detects colonization) • **GDH antigen + toxin A/B EIA** (multistep algorithm: GDH + toxin, arbitrated by NAAT; or NAAT + toxin) • toxin EIA alone not recommended (low sensitivity).
- **CT/abdominal imaging** if fulminant (megacolon, ileus) • lower endoscopy: pseudomembranes.

Severity	Criteria
Non-severe	**WBC ≤ 15,000 cells/mm³ and SCr < 1.5 mg/dL**
Severe	**WBC > 15,000 or SCr > 1.5 mg/dL**
Fulminant	**Hypotension or shock**, **ileus**, or **toxic megacolon** (± ↑ lactate, perforation)

Treatment

- **General: stop the inciting antibiotic** as soon as possible (or narrow it) • stop unnecessary **PPIs** • fluids/electrolytes • **avoid antimotility agents** (loperamide, opioids) in acute CDI (toxic megacolon) • start empiric treatment while awaiting results if high suspicion/severe.

Episode	Recommended treatment
Initial episode (non-severe or severe)	**Fidaxomicin 200 mg PO BID × 10 days** (**preferred** — lower recurrence) **or vancomycin 125 mg PO QID × 10 days • metronidazole 500 mg PO TID × 10–14 days** only for **non-severe** disease if fidaxomicin and vancomycin are **unavailable** (avoid repeated/prolonged courses — neurotoxicity)
Fulminant	**Vancomycin 500 mg PO/NG QID + metronidazole 500 mg IV q8h • + rectal vancomycin 500 mg in 100 mL NS q6h (retention enema)** if **ileus • early surgical consultation** (subtotal colectomy or diverting loop ileostomy with colonic lavage) • fluids, ICU care
First recurrence	**Fidaxomicin** (standard 200 mg BID × 10 days, **or extended-pulsed**: 200 mg BID × 5 days then 200 mg every other day days 7–25) — preferred • **or vancomycin taper and pulse** (125 mg QID × 10–14 days → BID × 7 days → daily × 7 days → every 2–3 days × 2–8 weeks) • **or standard vancomycin 125 mg QID × 10 days** if metronidazole was used for the first episode
Second or subsequent recurrence	**Fidaxomicin** (standard or extended-pulsed) • **vancomycin taper and pulse • vancomycin 125 mg QID × 10 days then rifaximin 400 mg TID × 20 days • fecal microbiota transplantation (FMT)** (after ≥ 2 recurrences, i.e., ≥ 3 episodes, with appropriate antibiotic treatment)
Prevention of recurrence (adjuncts)	**Live biotherapeutics** after completing CDI antibiotics in adults with recurrent CDI: **fecal microbiota, live-jslm (Rebyota)** — single **rectal** dose 24–72 h after antibiotics • **fecal microbiota spores, live-brpk (Vowst)** — **oral 4 capsules daily × 3 days** after **magnesium citrate** bowel prep, 2–4 days after antibiotics • **bezlotoxumab** (anti-toxin B antibody; HF caution) was recommended but has been **discontinued** (2025)

- **Oral vancomycin is NOT absorbed** (acts in colon) • **IV vancomycin does NOT treat CDI** (not excreted into the gut lumen) • compounded oral solution or capsules.
- **Fidaxomicin:** narrow-spectrum **macrocyclic**, minimally absorbed, spares normal flora → ↓ **recurrence** • expensive • ADRs: nausea, GI bleeding (rare), hypersensitivity.
- **Metronidazole:** absorbed systemically (reaches colon via inflamed mucosa) — **inferior** • ADRs: neuropathy, metallic taste, disulfiram-like reaction, ↑ INR.
- **Recurrence** occurs in ~20–30% after first episode (higher after each recurrence), usually within 8 weeks.
- **Probiotics are NOT recommended** for primary prevention or treatment (IDSA, ACG — insufficient evidence; risk of fungemia/bacteremia) • AGA suggests specific strains in some adults on antibiotics (conditional).

- **Secondary prophylaxis** with oral vancomycin 125 mg daily may be considered during subsequent antibiotic exposure in high-risk patients with prior CDI.
- **Children:** metronidazole or vancomycin (non-severe initial) • fidaxomicin ≥ 6 months • **don't test infants < 12 months** (colonization common).
- **Antimicrobial stewardship** = best prevention (↓ clindamycin, FQs, cephalosporins) • isolate patients, dedicated equipment.

13. C. difficile Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Fidaxomicin	**200 mg PO BID × 10 days** (or extended-pulsed)	Nausea, abdominal pain, GI bleed (rare)	**Preferred** • lowest recurrence • minimal absorption • cost
Vancomycin PO	**125 mg QID × 10 days** • fulminant 500 mg QID • taper/pulse for recurrence	Minimal (bitter taste)	**Oral only** (IV doesn't work) • not absorbed
Vancomycin rectal	500 mg in 100 mL NS q6h (retention enema)	Perforation risk	Fulminant with ileus
Metronidazole	**500 mg PO TID × 10–14 days** (fulminant 500 mg IV q8h)	**Neuropathy**, metallic taste, nausea	Only non-severe if others unavailable • disulfiram-like • ↑ INR
Rifaximin	400 mg TID × 20 days (after vancomycin)	Nausea, peripheral edema	Recurrent CDI chaser • minimal absorption
Fecal microbiota, live-jslm (Rebyota)	150 mL rectal single dose	Abdominal pain, diarrhea	Recurrent CDI prevention (after antibiotics)
Fecal microbiota spores, live-brpk (Vowst)	**4 capsules PO daily × 3 days** (after Mg citrate)	Bloating, fatigue	Recurrent CDI prevention • oral

Exam Traps and Frequently Tested Points

1. **IE organisms: S. aureus** most common (IDU → tricuspid) • viridans strep (dental) • enterococci (GU) • CoNS (early PVE) • **S. gallolyticus → colonoscopy** • HACEK.
2. **Osler nodes = painful** (immunologic) • **Janeway lesions = painless** (embolic) • Roth spots, splinter hemorrhages, glomerulonephritis.
3. **Duke definite:** 2 major, 1 major + 3 minor, or 5 minor • **3 sets of blood cultures before antibiotics** • TTE first → TEE (PVE, S. aureus, abscess).
4. **Duration from first negative blood culture** • NVE ~4 weeks (strep) / 6 weeks (staph, enterococcus) • PVE ≥ 6 weeks.
5. **Strep IE:** penicillin G or ceftriaxone 2 g × 4 weeks (or 2 weeks + gentamicin) • vancomycin if allergic.
6. **MSSA IE:** nafcillin/oxacillin (or cefazolin) × 6 weeks • **no gentamicin for NVE** • **MRSA:** vancomycin or daptomycin ≥ 8 mg/kg • β-lactam beats vancomycin for MSSA.
7. **PVE staph: add rifampin (biofilm) + gentamicin first 2 weeks** • start rifampin after bacteremia clears.
8. **Enterococcal IE: ampicillin + ceftriaxone** (CrCl < 50 or HLAR) or ampicillin + gentamicin • **cephalosporins alone don't cover enterococci**.
9. **Gentamicin synergy 3 mg/kg/day** (peak 3–4, trough < 1) • **daptomycin**: CPK weekly, not for pneumonia • nafcillin: interstitial nephritis, ↓ warfarin.
10. **Surgery:** HF, abscess/heart block, persistent bacteremia, fungal, vegetation > 10 mm with emboli • no aspirin for IE.
11. **IE prophylaxis:** only highest-risk (prosthetic valve, prior IE, certain CHD, transplant valvulopathy) + **dental procedures** • **amoxicillin 2 g** 30–60 min before • allergy: azithromycin 500 mg, doxycycline, cephalexin • **no clindamycin** • not for GI/GU.
12. **SSTI: purulent = S. aureus** (I&D ± TMP-SMX/doxy) • **non-purulent = strep** (cephalexin, penicillin, dicloxacillin, clindamycin × 5 days) • elevate limb, treat tinea pedis.

13. **TMP-SMX and doxycycline cover MRSA but poor/unreliable strep • cephalexin/dicloxacillin don't cover MRSA** • clindamycin covers both (resistance; C. diff).
14. **Necrotizing fasciitis:** pain out of proportion → **surgery** + vancomycin + pip-tazo + **clindamycin (toxin)** • GAS → penicillin + clindamycin • Vibrio → doxy + ceftazidime.
15. **Bites: amoxicillin-clavulanate** (Pasteurella — cats; Eikenella — humans) • avoid cephalexin/dicloxacillin/clindamycin alone • tetanus, rabies.
16. **Diabetic foot:** mild → gram-positive (cephalexin, amox-clav; MRSA → doxy/TMP-SMX) 1–2 weeks • moderate–severe → polymicrobial (amox-clav, ertapenem, pip-tazo if Pseudomonas risk) • osteomyelitis 6 weeks • deep tissue cultures, offloading.
17. **Osteomyelitis:** S. aureus most common • **Salmonella (sickle cell)** • **Pseudomonas (sneaker puncture)** • MRI + bone biopsy • **6 weeks** • oral FQs, linezolid, TMP-SMX OK.
18. **Septic arthritis:** arthrocentesis • S. aureus → vancomycin • gonococcal → ceftriaxone 1 g daily • rifampin combination for prosthetic hardware.
19. **Meningitis triad** fever + neck stiffness + altered mental status • **antibiotics within 1 h, don't wait for CT/LP** • CT first if immunocompromised, seizure, papilledema, focal deficit, ↓ consciousness.
20. **CSF bacterial:** ↑ PMNs, **↓ glucose (< 40)**, ↑↑ protein • viral: lymphocytes, normal glucose.
21. **Empiric meningitis:** < 1 month **ampicillin + cefotaxime (no ceftriaxone in neonates)** • 1 month–50 years **vancomycin + ceftriaxone** • **> 50 years/immunocompromised + ampicillin (Listeria)** • post-neurosurgery vancomycin + cefepime/meropenem.
22. **Ceftriaxone 2 g q12h** (meningitic dose) • **ampicillin 2 g q4h** • **cephalosporins don't cover Listeria** (TMP-SMX in PCN allergy).
23. **Dexamethasone 0.15 mg/kg q6h × 2–4 days before/with first antibiotic** → benefit in pneumococcal (adults) and Hib (children) • stop if other organism • ↓ vanco CSF penetration.
24. **Duration:** meningococcus/Hib 7 days • pneumococcus 10–14 • GBS 14–21 • **Listeria ≥ 21** • gram-negatives 21 days.
25. **Meningococcal prophylaxis: rifampin 600 mg q12h × 2 days**, **ciprofloxacin 500 mg once**, or **ceftriaxone 250 mg IM once (pregnancy)** • droplet precautions 24 h • Hib prophylaxis rifampin × 4 days.
26. **HSV encephalitis:** temporal lobe • **acyclovir 10 mg/kg q8h (IBW) × 14–21 days** • hydrate (crystal nephropathy) • cryptococcal: L-AmB + flucytosine → fluconazole, serial LPs.
27. **IAI = source control + antibiotics** • CA mild–moderate: **ceftriaxone/cefazolin + metronidazole**, ertapenem, FQ + metronidazole • severe/HA: **pip-tazo, carbapenem, cefepime + metronidazole**.
28. **Avoid ampicillin-sulbactam (E. coli), cefotetan and clindamycin (B. fragilis)** for IAI • no routine antifungal/enterococcal coverage in community-acquired.
29. **IAI duration 4 days after source control** (STOP-IT) • appendectomy for uncomplicated appendicitis → ≤ 24 h • uncomplicated diverticulitis may not need antibiotics.
30. **SBP:** cefotaxime 2 g q8h or ceftriaxone × 5–7 d + **albumin 1.5 g/kg day 1, 1 g/kg day 3** • prophylaxis cipro or TMP-SMX • necrotizing pancreatitis: no prophylactic antibiotics.
31. **Ertapenem** lacks Pseudomonas, Acinetobacter, enterococcus • **carbapenems ↓ valproate** • **metronidazole**: disulfiram-like, ↑ INR, neuropathy • tigecycline ↑ mortality.
32. **CDI risk:** clindamycin, FQs, cephalosporins, carbapenems, PPIs, age, hospitalization • **alcohol sanitizer doesn't kill spores → soap and water**, bleach, contact precautions.
33. **CDI testing:** only ≥ 3 unformed stools/24 h, no laxatives in 48 h • **no test of cure**, don't test asymptomatic.
34. **CDI severity:** non-severe WBC ≤ 15,000 and SCr < 1.5 • severe WBC > 15,000 or SCr > 1.5 • **fulminant** hypotension/shock, ileus, megacolon.
35. **CDI initial: fidaxomicin 200 mg BID × 10 d (preferred)** or **vancomycin 125 mg PO QID × 10 d** • metronidazole only if others unavailable (non-severe).
36. **Fulminant CDI:** vancomycin 500 mg PO/NG QID + metronidazole 500 mg IV q8h (+ rectal vancomycin if ileus) + surgery consult.
37. **Recurrence:** fidaxomicin or vancomycin taper/pulse • ≥ 2 recurrences → FMT, vancomycin → rifaximin • Vowst (oral) / Rebyota (rectal) after antibiotics • **IV vancomycin doesn't work** • avoid loperamide • stop PPIs • probiotics not recommended.

Note: doses and recommendations follow current AHA/ESC, IDSA, SIS, ACG and SHEA guidance. If your course slides or reference use slightly different numbers, **use your course numbers for the exam**.